Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 5 / 12

Worked Example: The Uniform Distribution and the Failure of Regularity

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Contents

1 Worked example: $\mathrm{U}[0, \theta]$

The uniform family $\mathrm{U}[0, \theta]$ is a one-parameter family that contrasts sharply with the normal. It is the simplest *non-regular* family: the support $[0, \theta]$ depends on the parameter. This breaks the score identity $\mathbb{E}[s] = 0$, and with it the CLT for the score and the usual asymptotic theory.

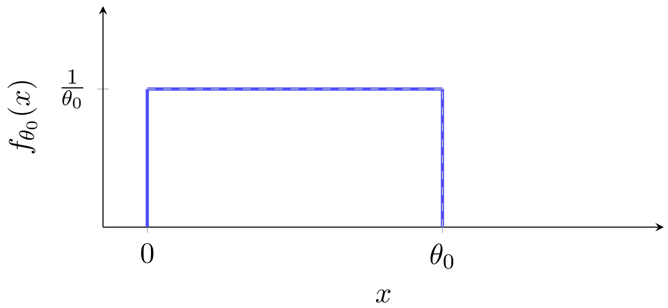
1.1 Step 1. Density family

For a fixed true parameter $\theta_0 > 0$, the density is

$$f_{\theta_0}(x) \;=\; \frac{1}{\theta_0}\,\mathbf{1}_{x\in[0,\theta_0]}(x).$$

A flat rectangle of height $1/\theta_0$ over $[0, \theta_0]$.

Density $f_{\theta_0}(x) = \frac{1}{\theta_0}\mathbf{1}_{[0,\theta_0]}$

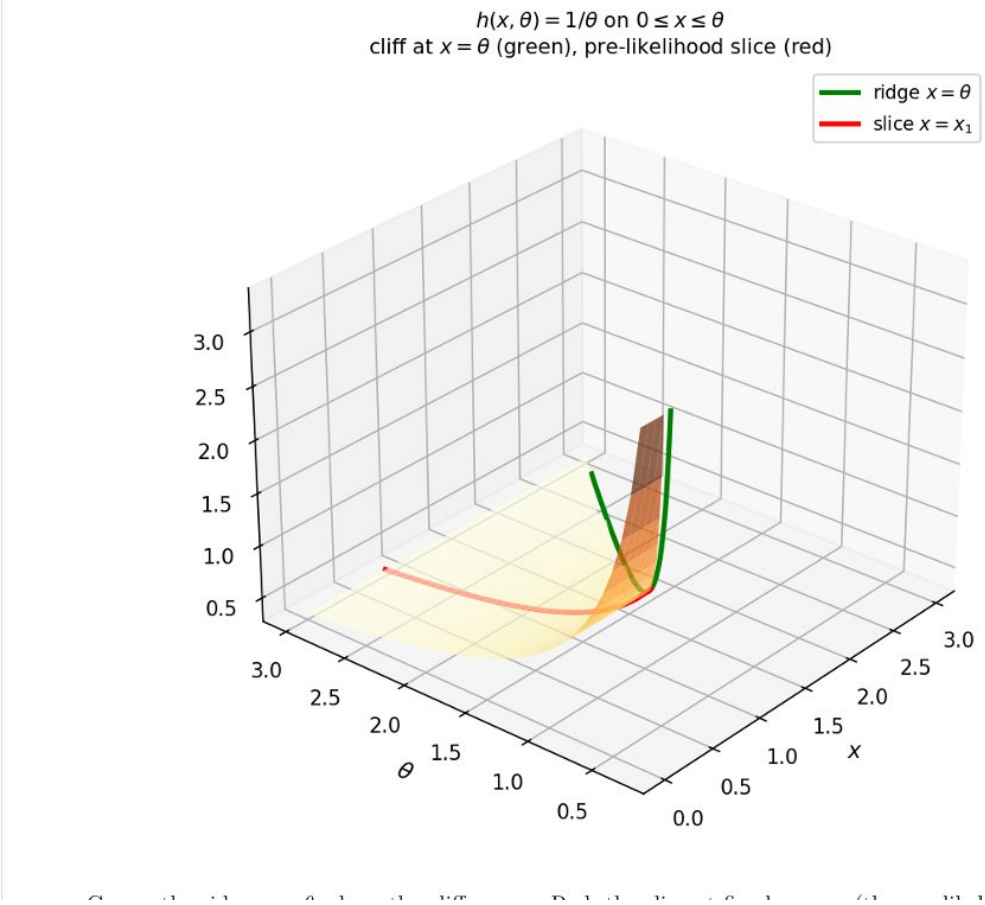


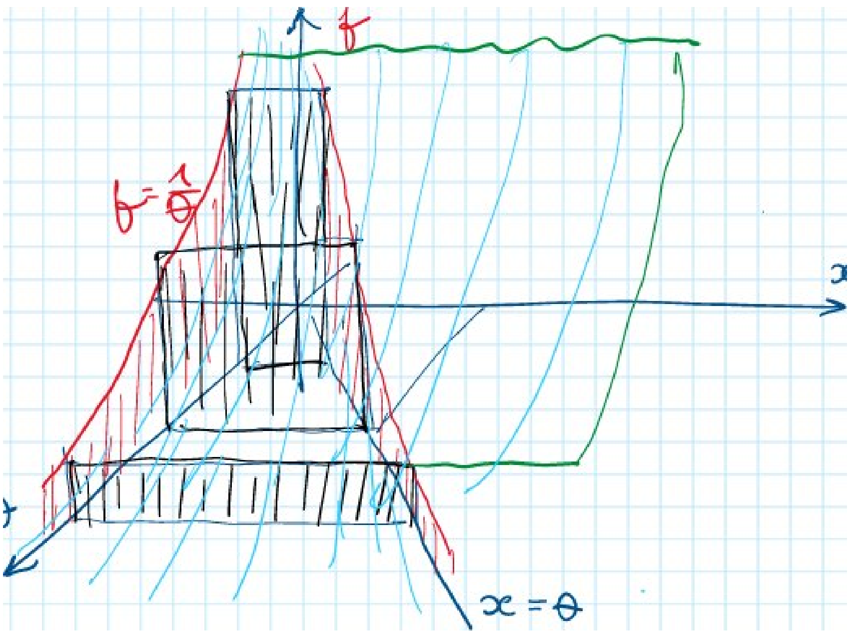
1.2 Step 2. Bivariate surface $h(x, \theta)$

The density as a function of both arguments is

$$h(x,\theta) \;=\; \frac{1}{\theta}\,\mathbf{1}_{x\in[0,\theta]}.$$

The surface lives above the triangular region $\{0 \le x \le \theta\}$, where it takes the value $1/\theta$ (flat in x, decreasing in θ). It drops to zero beyond the boundary $x = \theta$. The boundary $x = \theta$ is the ridge (cf. the normal case), but here the surface has a *cliff* rather than a smooth peak.





Green: the ridge $x = \theta$ where the cliff occurs. Red: the slice at fixed $x = x_1$ (the pre-likelihood).

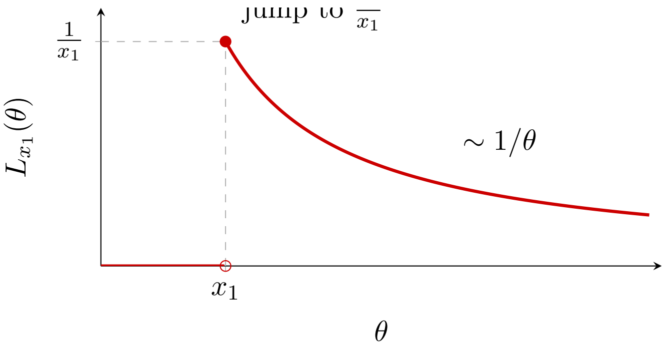
1.3 Step 3. Pre-likelihood

Fix $x = x_1$ and vary θ. The pre-likelihood is

$$L_{x_1}(\theta) \;=\; h(x_1, \theta) \;=\; \frac{1}{\theta}\,\mathbf{1}_{\theta \ge x_1}.$$

For $\theta < x_1$ the observation x_1 lies outside the support $[0, \theta]$, so the density is zero. At $\theta = x_1$ the density jumps to $1/x_1$, then decays as $1/\theta$ for $\theta > x_1$.

Pre-likelihood $L_{x_1}(\theta) = \frac{1}{\theta}\mathbf{1}_{\theta \ge x_1}$



The pre-likelihood is maximised at its *left boundary* $\theta = x_1$: since $1/\theta$ is decreasing, the largest value is achieved at the smallest admissible θ. Hence the MLE from a single observation is

$$\hat{\theta} \;=\; x_1.$$

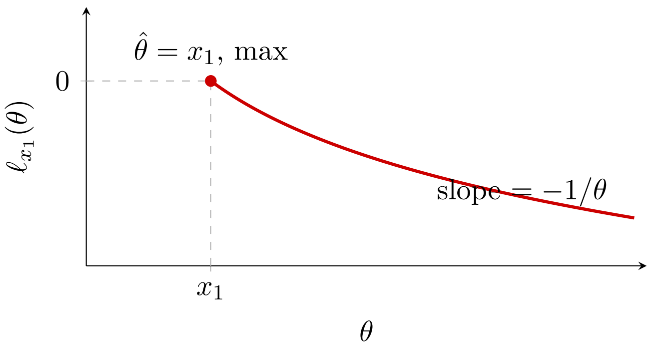
1.4 Step 4. Log-pre-likelihood

Taking logs on the support $\theta \ge x_1$:

$$\ell_{x_1}(\theta) \;=\; \ln h(x_1, \theta) \;=\; \ln \frac{1}{\theta} \;=\; -\ln\theta, \qquad \theta \ge x_1.$$

Outside this region $\ell_{x_1}(\theta) = -\infty$ (or undefined). The function $-\ln\theta$ is strictly *decreasing*, so its maximum is again at the left boundary $\theta = x_1$.

Log-pre-likelihood $\ell_{x_1}(\theta) = -\ln\theta$**,** $\theta \ge x_1$ **(undefined for** $\theta < x_1$**)**



1.5 Step 5. n observations: joint likelihood and MLE

For an i.i.d. sample $(x_1, \dots, x_n)$ the joint density is

$$h(x_1, \dots, x_n; \theta) \;=\; \prod_{i=1}^{n} \frac{1}{\theta}\,\mathbf{1}_{x_i \le \theta} \;=\; \frac{1}{\theta^n}\,\mathbf{1}_{\theta \ge x_{(n)}},$$

where $x_{(n)} = \max_i x_i$. The constraint $\theta \ge x_i$ for all i collapses to $\theta \ge x_{(n)}$.

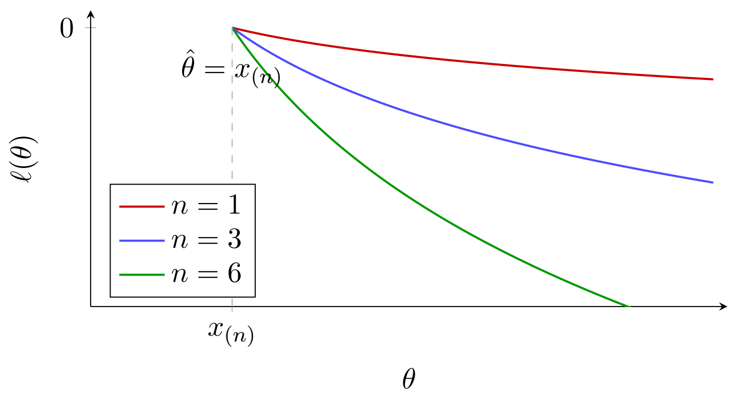
The joint log-likelihood is

$$\ell(\theta) \;=\; -n\ln\theta, \qquad \theta \ge x_{(n)}.$$

Again a decreasing function, maximised at the left boundary:

$$\boxed{\hat{\theta} \;=\; x_{(n)} \;=\; \max(x_1, \dots, x_n).}$$

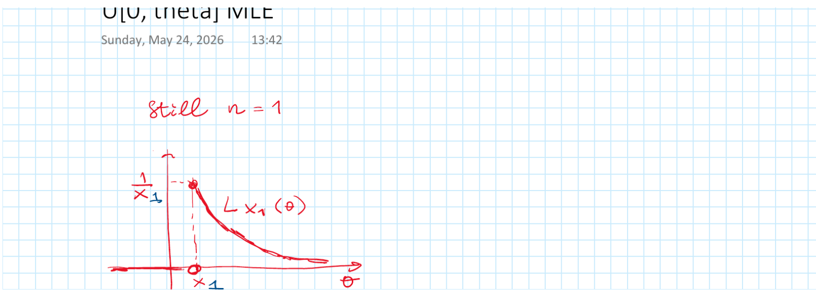
Joint log-likelihood $\ell(\theta) = -n \ln \theta$ for $n = 1, 3, 6$; boundary at $x_{(n)} = 1$



As n grows the log-likelihood steepens: each additional observation tightens the constraint on θ from below, and the function descends faster.

1.6 Step 5a. MLE: explicit derivation and consistency

$n = 1$: MLE from the pre-likelihood. Fix $x_1 > 0$. The pre-likelihood $L_{x_1}(\theta) = \frac{1}{\theta} \mathbf{1}_{x_1 \le \theta}$ is a function of θ with parameter x_1, always non-negative.

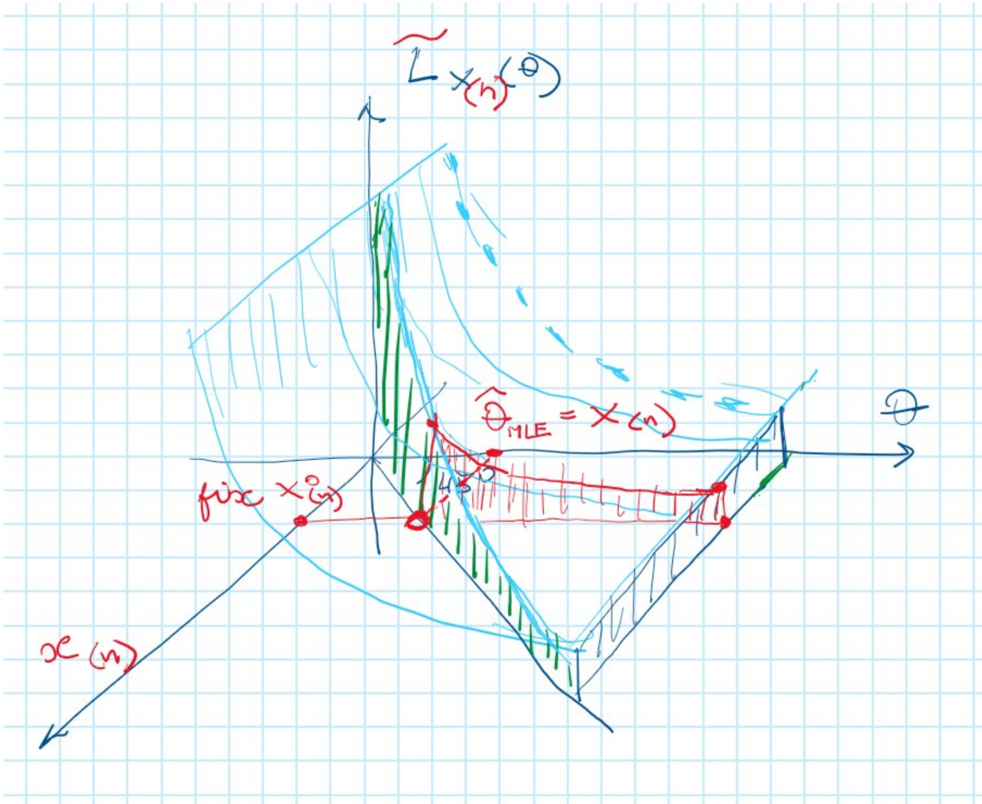


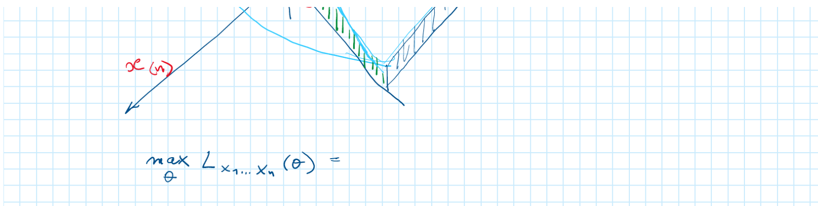
On the feasible region $\theta \ge x_1$ the function $1/\theta$ is strictly decreasing. Therefore the maximum is attained at the left boundary:

$$\max_{\theta} L_{x_1}(\theta) \;=\; \frac{1}{x_1} \;=\; L_{x_1}(\theta)\big|_{\theta = x_1}, \qquad \operatorname{argmax}_{\theta} L_{x_1}(\theta) \;=\; x_1.$$

n observations: sufficiency and the 3D picture. For n i.i.d. observations the likelihood reduces to $\widetilde{L}_{x_{(n)}}(\theta) = \theta^{-n} \mathbf{1}_{\theta \ge x_{(n)}}$. The same argument applies: on $[x_{(n)}, \infty)$ the function θ^{-n} is strictly decreasing, so the maximum is at $\theta = x_{(n)}$.

The Probabilist's 3D picture shows the surface $\widetilde{L}_{x_{(n)}}(\theta)$ with the cliff at $\theta = x_{(n)}$ and the MLE marked on the θ-axis:





The full maximisation chain:

$$\max_\theta L_{x_1,\dots,x_n}(\theta) \;=\; \max_\theta \widetilde{L}_{x_{(n)}}(\theta) \;=\; \widetilde{L}_{x_{(n)}}(\theta)\big|_{\theta = x_{(n)}},$$

$$\hat{\theta}_{\text{MLE}}(x_1,\dots,x_n) \;=\; \text{argmax}_\theta\, L_{x_1,\dots,x_n}(\theta) \;=\; x_{(n)}(x_1,\dots,x_n).$$

Consistency: $x_{(n)} \xrightarrow{p} \theta_0$. *Goal.* Assume $\theta = \theta_0$ is the true parameter and $(x_1,\dots,x_n)$ is an i.i.d. $\mathrm{U}[0,\theta_0]$ sample. We want to show

$$X_{(n)} = \max_{i=1,\dots,n} X_i \xrightarrow{p} \theta_0 \quad \text{as } n \to \infty.$$

By definition this means: for every $\varepsilon > 0$, $P\{|X_{(n)} - \theta_0| > \varepsilon\} \to 0$.
Proof. Fix $\varepsilon > 0$. Decompose the event:

$$\{|X_{(n)} - \theta_0| > \varepsilon\} = \{X_{(n)} > \theta_0 + \varepsilon\} \cup \{X_{(n)} < \theta_0 - \varepsilon\}.$$

The first event is *empty* by assumption ($X_i \le \theta_0$ always, so $X_{(n)} \le \theta_0 < \theta_0 + \varepsilon$). Therefore:

$$\{|X_{(n)} - \theta_0| > \varepsilon\} = \{X_{(n)} < \theta_0 - \varepsilon\} = \bigcap_{i=1}^n \{X_i < \theta_0 - \varepsilon\}.$$

Taking probabilities and using independence then identical distribution:

$$P\{|X_{(n)} - \theta_0| > \varepsilon\} = P\left\{\bigcap_{i=1}^{n}\{X_i < \theta_0 - \varepsilon\}\right\} \overset{\text{ind.}}{=} \prod_{i=1}^{n} P\{X_i < \theta_0 - \varepsilon\}$$
$$\overset{\text{i.d.}}{=} \Big(P\{X_1 < \theta_0 - \varepsilon\}\Big)^n \overset{X\sim\mathrm{U}[0,\theta_0]}{=} \left(\frac{\theta_0 - \varepsilon}{\theta_0}\right)^n \longrightarrow 0,$$

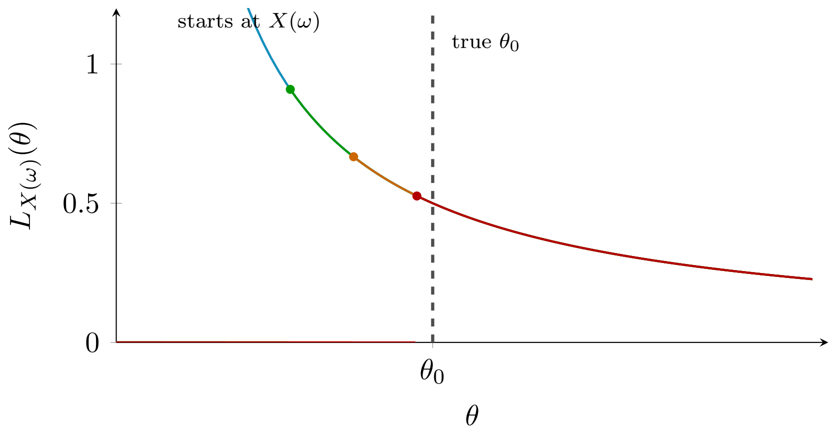
because $\varepsilon > 0$ implies $(\theta_0 - \varepsilon)/\theta_0 < 1$. □

Remark 1 (Almost sure convergence). In fact $X_{(n)} \xrightarrow{a.s.} \theta_0$ as well, because $X_{(n)}$ is monotone non-decreasing in n: adding an observation can only increase the maximum. A bounded monotone sequence converges almost surely.

1.7 Step 6. Random curves

Treat the data as random: $X(\omega) \sim \mathrm{U}[0, \theta_0]$. Each ω produces one observation $X(\omega) \in [0, \theta_0]$, and hence one pre-likelihood curve $(1/\theta) \cdot \mathbf{1}_{\theta \geq X(\omega)}$. Smaller observations permit smaller θ (curves start earlier); larger observations push the admissible region to the right. For $n = 1$, since $X \sim \mathrm{U}[0, \theta_0]$, the single observation is uniform, so the starting points of the individual curves are indeed uniformly distributed on $[0, \theta_0]$.

Cloud of random pre-likelihood curves, $\theta_0 = 2$ (true value, dashed)



All curves agree that $\theta \geq x_{\max} = \max_i X_i(\omega)$, and each votes for the smallest possible θ. The MLE $\hat{\theta} = x_{(n)}$ is the smallest θ consistent with all observed data.

Remark 2 (Distribution of the MLE: order statistics and the Beta). For $n > 1$ the starting point of the *joint* likelihood is $X_{(n)} = \max(X_1, \ldots, X_n)$, not a single uniform observation. For i.i.d. $X_i \sim \mathrm{U}[0, \theta_0]$ the k-th order statistic has the scaled Beta distribution:

$$X_{(k)} \;\sim\; \theta_0 \cdot \mathrm{Beta}(k,\, n-k+1), \qquad f_{X_{(k)}}(x) \;=\; \frac{n!}{(k-1)!(n-k)!}\left(\frac{x}{\theta_0}\right)^{k-1}\left(1 - \frac{x}{\theta_0}\right)^{n-k}\frac{1}{\theta_0}.$$

In particular the MLE $\hat{\theta} = X_{(n)} \sim \theta_0 \cdot \mathrm{Beta}(n, 1)$ has density $f(x) = nx^{n-1}/\theta_0^n$ on $[0, \theta_0]$: it is *right-skewed toward* θ_0, not uniform. Its mean and bias are

$$\mathbb{E}[\hat{\theta}] \;=\; \frac{n}{n+1}\theta_0, \qquad \text{Bias} \;=\; -\frac{\theta_0}{n+1} \;\to\; 0.$$

The bias-corrected estimator $\frac{n+1}{n} X_{(n)}$ is unbiased. The deviation $\theta_0 - X_{(n)}$ is $O(1/n)$ in probability, which explains why the MLE converges at rate $1/n$ rather than $1/\sqrt{n}$: each new observation can only shrink the gap $[\,X_{(n)}, \theta_0\,]$, and after n uniform draws the expected gap is $\theta_0/(n+1)$.

The minimum $X_{(1)} \sim \theta_0 \cdot \mathrm{Beta}(1, n)$ is left-skewed toward 0, with mean $\theta_0/(n+1)$. The full picture: the k-th order statistic has mean $k\theta_0/(n+1)$, so the n order statistics divide $[0, \theta_0]$ into $n+1$ equal-mean spacings — a classical result on uniform spacings.

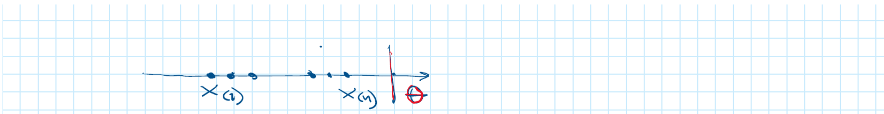
1.8 Step 5b. Multiple observations: geometry of the joint likelihood

The joint likelihood $L_{x_1,\ldots,x_n}(\theta)$ factorises over the i.i.d. sample and simplifies to a function of the order statistic $x_{(n)} = \max_i x_i$ alone:

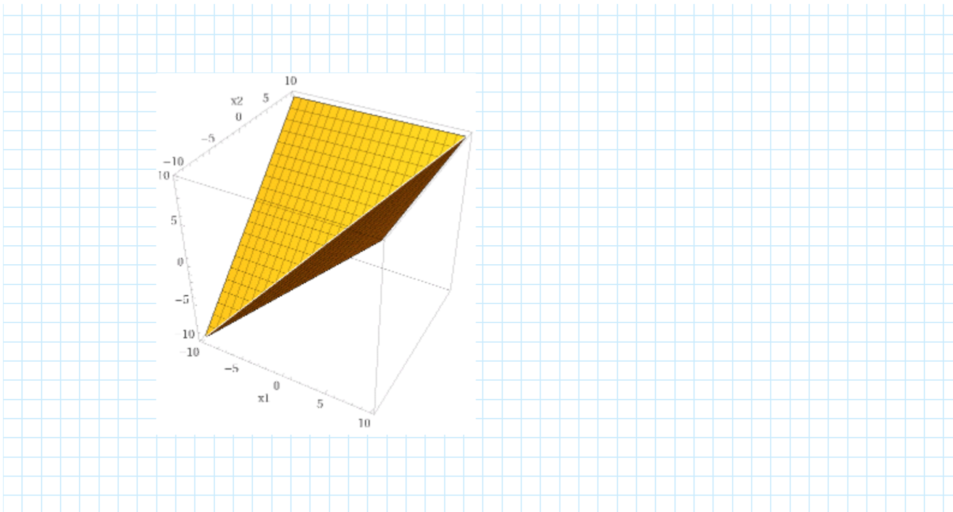
$$L_{x_1,\ldots,x_n}(\theta) \;=\; \prod_{i=1}^{n} \frac{1}{\theta}\,\mathbf{1}_{x_i \le \theta} \;=\; \frac{1}{\theta^n}\,\mathbf{1}_{x_{(n)} \le \theta}.$$

The key reduction step: all n constraints $x_i \le \theta$ collapse into the single constraint $\max(x_1, \ldots, x_n) \le \theta$.

Number line picture. On the θ-axis, the data $x_{(1)} \le \cdots \le x_{(n)}$ cluster to the left of θ; the constraint forces θ to the right of the largest observation.

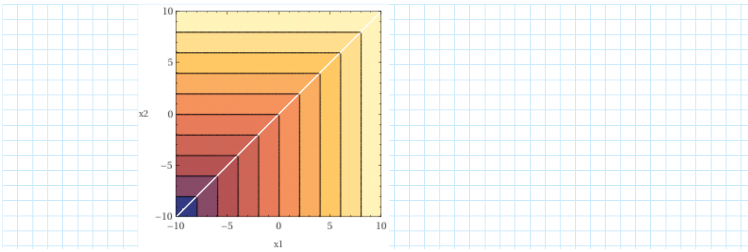


The joint likelihood as a 3D surface for $n = 2$. For $n = 2$ fix θ and let (x_1, x_2) vary. The constraint region $\{(x_1, x_2) : x_1 \le \theta,\, x_2 \le \theta\}$ is the square $[0, \theta]^2$; the likelihood equals $1/\theta^2$ inside and 0 outside, forming a wedge-shaped surface above the positive quadrant.



As θ increases the square grows and the height $1/\theta^2$ decreases: more parameter space is consistent with the data, but each point has lower density.

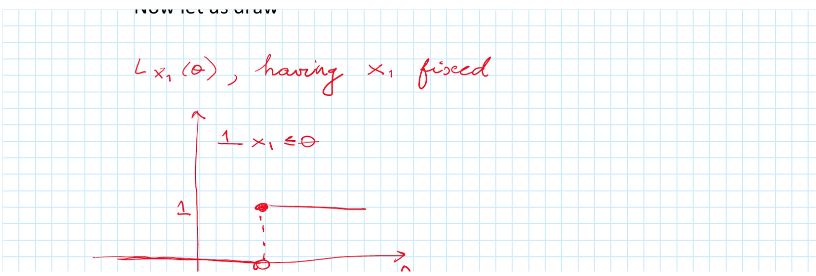
Contour view. The level sets of $h(x_1, x_2; \theta)$ as a function of (x_1, x_2) for fixed θ are squares with the corner at the origin: brighter (yellow) = larger θ = lower density; darker (red/purple) = smaller θ = higher density. The white diagonal $x_1 = x_2$ marks the ridge.



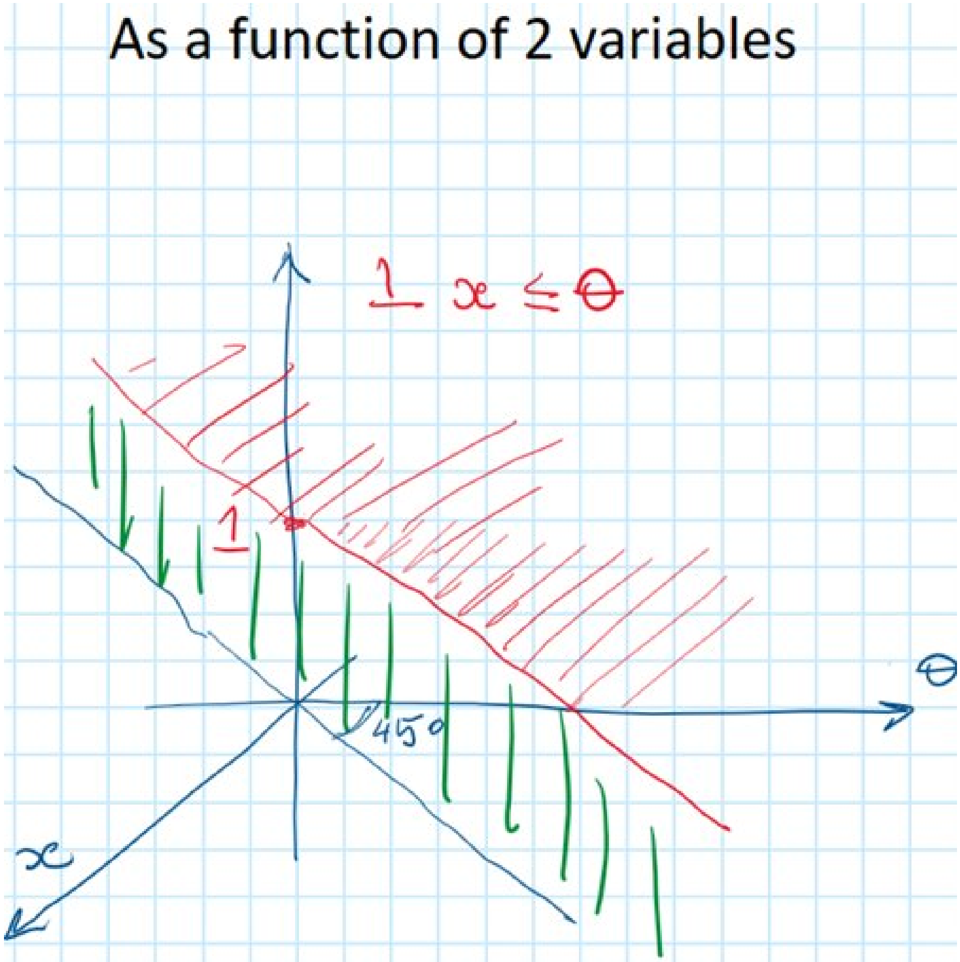
Building $L_{x_1}(\theta)$ from two factors. For $n = 1$ fix x_1 and vary θ. The pre-likelihood $L_{x_1}(\theta)$ is the product of two bivariate functions:

1. The **indicator $\mathbf{1}_{x\leq\theta}$**: flat value 1 on the region $x \leq \theta$ (above the diagonal), zero below.
2. The **hyperbolic factor** $1/\theta$: a surface that depends only on θ, declining toward the θ-axis.

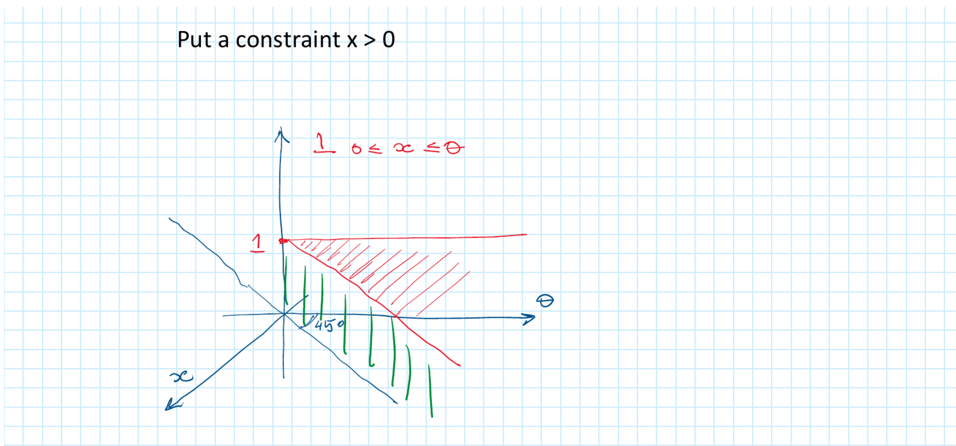
Step 1: $L_{x_1}(\theta)$ for fixed x_1. The 2D section at fixed x_1 shows the cliff structure: zero for $\theta < x_1$, jump to $1/x_1$ at $\theta = x_1$, then $1/\theta$ decay.



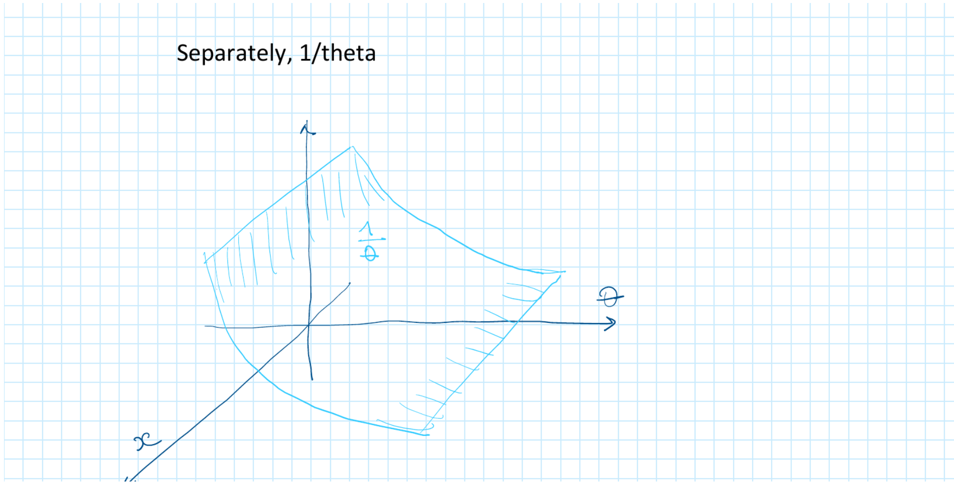
Step 2: $\mathbf{1}_{x\leq\theta}$ as a bivariate surface. Viewed as a function of both x and θ, the indicator is flat ($= 1$) over the triangular region below the diagonal $x = \theta$, with a vertical cliff on the ridge. Green curtains mark the contribution at fixed θ; red hatching shows the domain constraint.



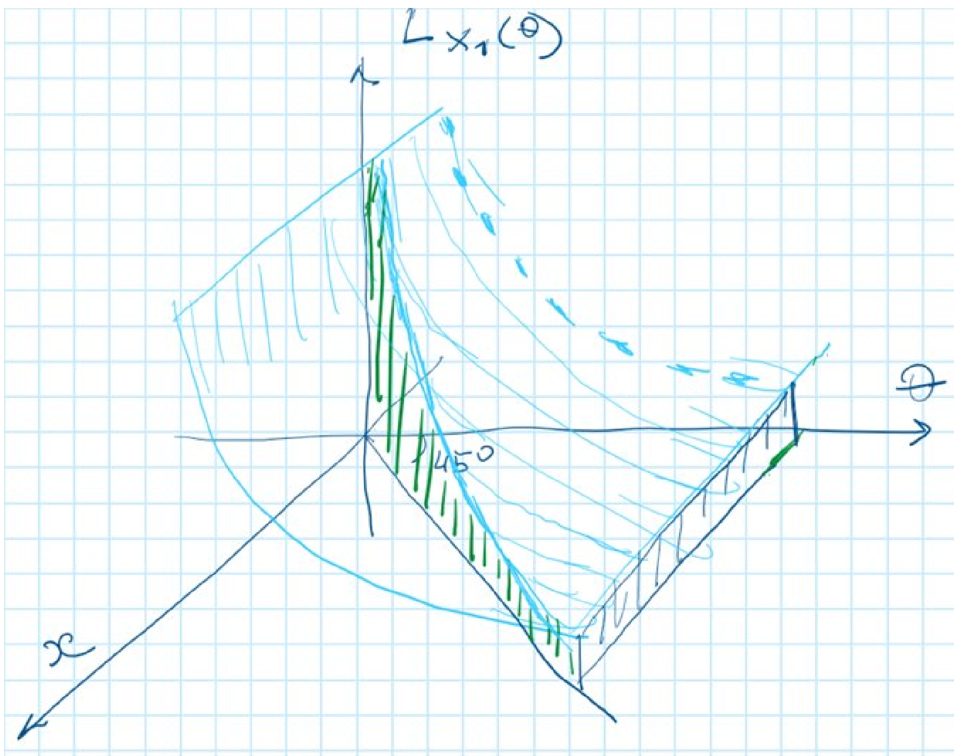
Step 3: Adding the constraint $x > 0$. The indicator $\mathbf{1}_{0 \le x \le \theta}$ restricts to $x \ge 0$, cutting out the left half-plane. The resulting surface lives strictly in the first quadrant.



Step 4: The factor $1/\theta$ *separately.* Viewed as a surface over the (x, θ) plane, $1/\theta$ is a "scroll": flat in x, hyperbolic in θ, declining away from the θ-axis.



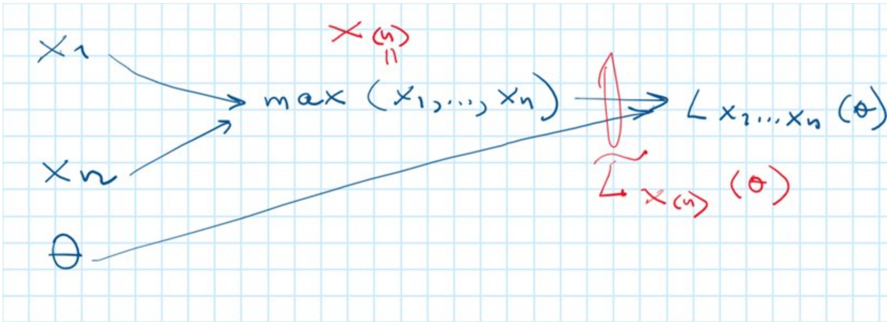
Step 5: The product $= L_{x_1}(\theta)$ *surface.* Multiplying the constraint surface by $1/\theta$ gives the full bivariate surface $h(x,\theta) = (1/\theta) \cdot \mathbf{1}_{0\le x\le\theta}$: the cliff at $x = \theta$, zero for $x > \theta$, and $1/\theta$ decay for $x < \theta$.



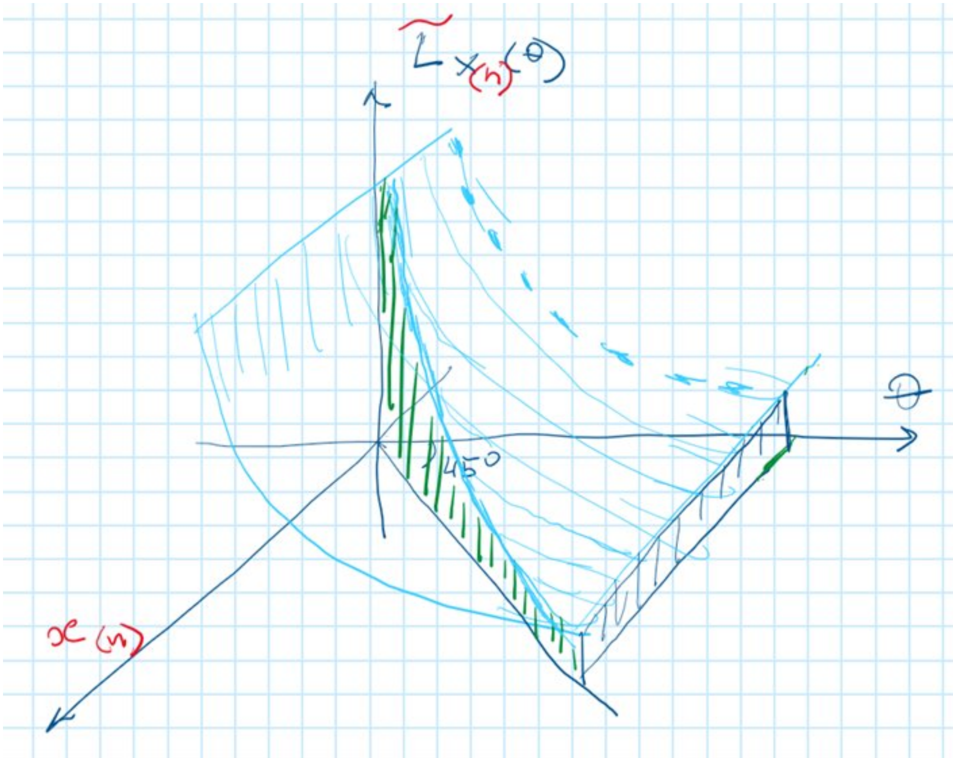
Sufficiency: the data collapse to $x_{(n)}$. The key structural insight is that the n-dimensional data vector $(x_1, \ldots, x_n)$ enters the likelihood *only* through $x_{(n)} = \max_i x_i$. The Probabilist's diagram below shows the chain:

$$(x_1, \ldots, x_n) \xrightarrow{\max} x_{(n)} \longrightarrow L_{x_1,\ldots,x_n}(\theta) = \widetilde{L}_{x_{(n)}}(\theta).$$

This is a simple example of **sufficiency**: $x_{(n)}$ is a sufficient statistic for θ in the $\mathrm{U}[0, \theta]$ family.



Once we know $x_{(n)}$, the likelihood is fully determined. The 3D geometry collapses: the full n-dimensional surface reduces to the same cliff-and-decay shape we saw for $n = 1$, but now with the cliff at $\theta = x_{(n)}$ instead of x_1.



Remark 3 (Homework: verify $x_{(n)}$ is MLE). Check directly that $\hat{\theta} = x_{(n)}$ maximises $L_{x_1,\dots,x_n}(\theta) = \theta^{-n}\mathbf{1}_{\theta \geq x_{(n)}}$ over $\theta > 0$. (Hint: the function is strictly decreasing in θ on the feasible region $[\,x_{(n)}, \infty)$.)

Next: check consistency of $\hat{\theta} = x_{(n)}$, i.e., that $x_{(n)} \xrightarrow{p} \theta_0$ as $n \to \infty$.

1.9 Step 4b. Log-pre-likelihood: extended treatment

The log-pre-likelihood for a single observation x_1 is

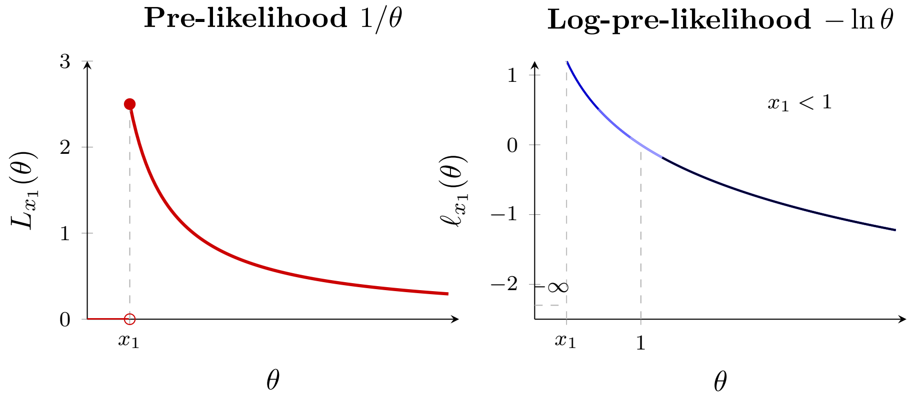
$$\ell_{x_1}(\theta) \;=\; \ln h(x_1, \theta) \;=\; \begin{cases} -\infty, & \theta < x_1, \\ -\ln\theta, & \theta \geq x_1, \end{cases}$$

which can be written compactly (abusing notation, $(-\infty) \cdot 0 := 0$) as

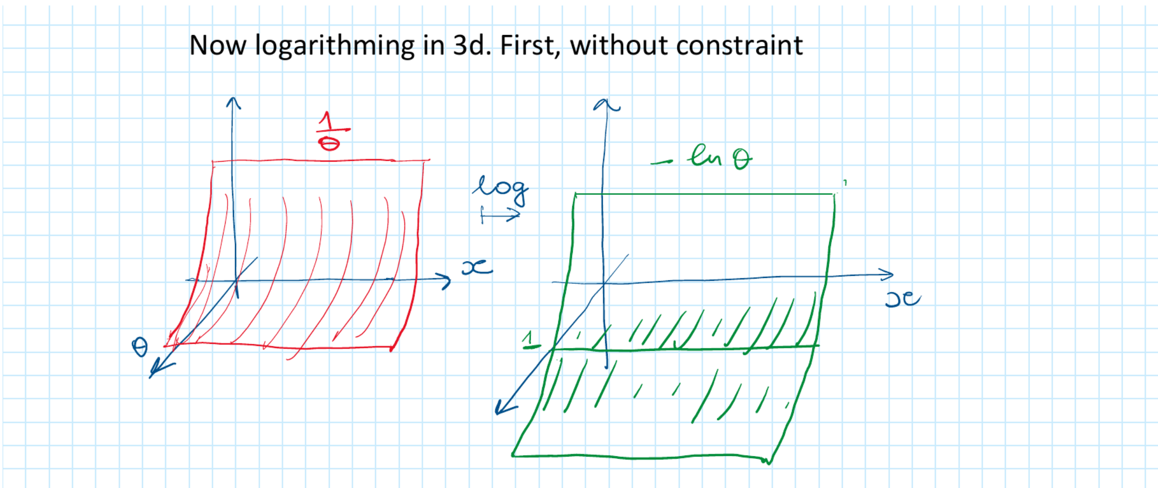
$$\ell_{x_1}(\theta) \;=\; (-\infty) \cdot \mathbf{1}_{\theta < x_1} \;+\; (-\ln\theta) \cdot \mathbf{1}_{\theta \geq x_1}.$$

The MLE is therefore $\hat{\theta}_{\text{MLE}} = \operatorname{argmax}_\theta \ell_{x_1}(\theta) = x_1$.

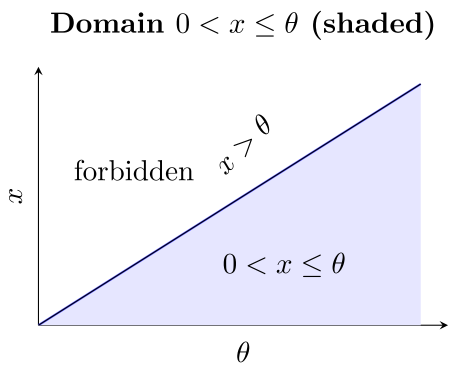
The figure below shows (left) the pre-likelihood $1/\theta$ in red and (right) the log-pre-likelihood $-\ln\theta$ in blue, for several values of $x_1 < 1$. Each curve starts at its own x_1 and the log-curves all pass through the value $-\ln x_1$ at the jump.



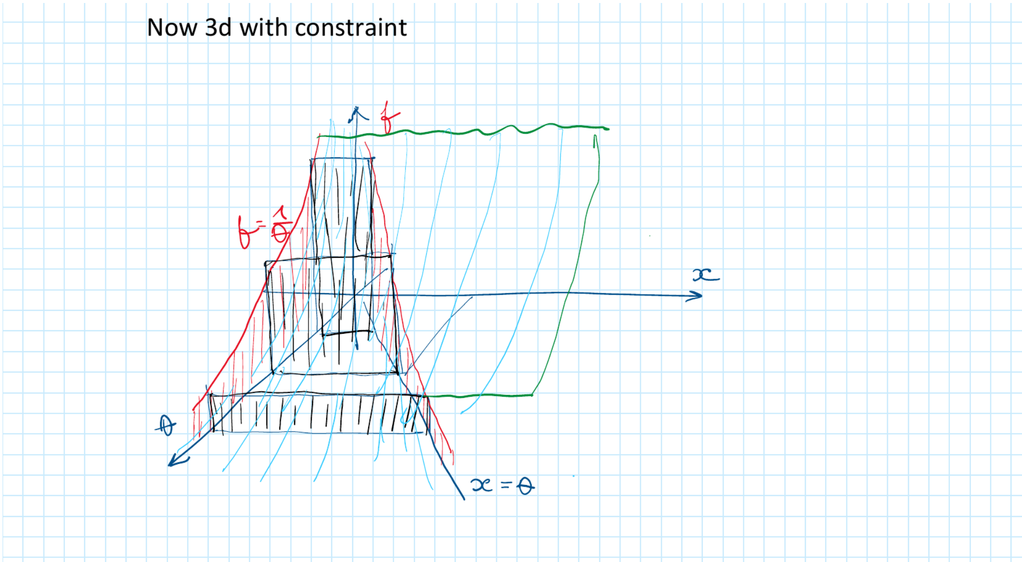
The bivariate log-likelihood surface. The function $\ell(x,\theta) = -\ln\theta$ (ignoring the constraint) is a surface that depends only on θ: flat in x, decreasing in θ. Without the constraint it looks like a rectangular slab tilted toward $\theta \to \infty$:



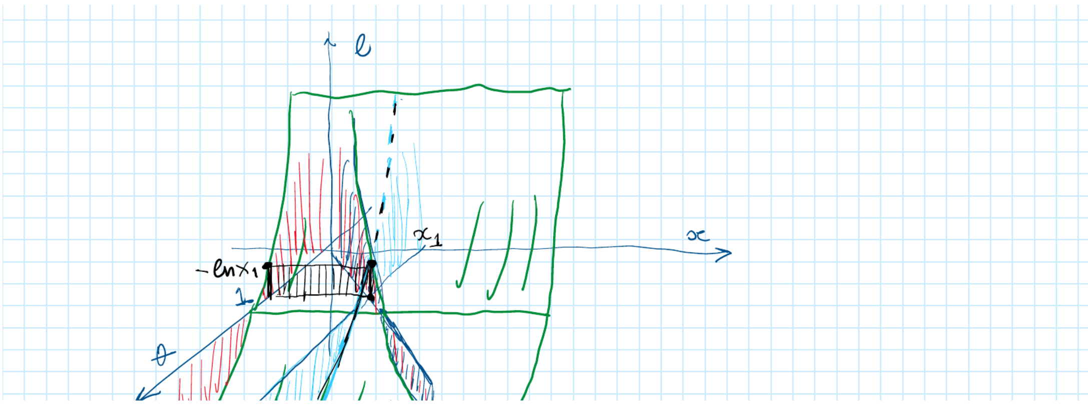
The domain constraint $0 < x \le \theta$ cuts out the triangular region below the diagonal $x = \theta$:



Adding the constraint gives the full 3D log-likelihood surface restricted to $0 < x \le \theta$ (the wedge below the diagonal):



Cutting this surface with the vertical plane $x = x_1$ (the x_1-section) extracts exactly the log-pre-likelihood $-\ln\theta$ for $\theta \geq x_1$ shown in the 2D plot above. That section is visible as the black curtain in the figure below:



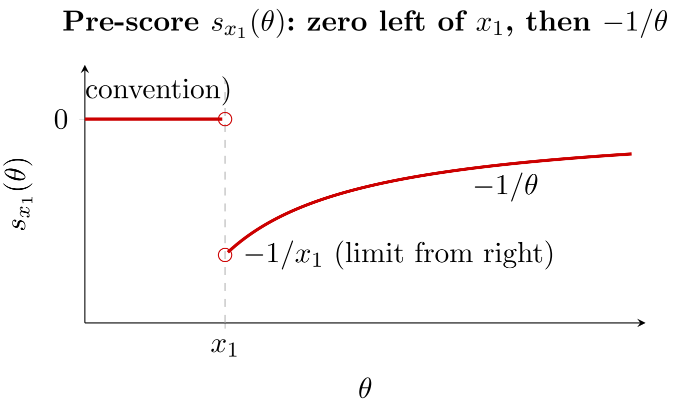
1.10 Step 7b. Pre-score: 2D and 3D geometry

The pre-score is the θ-derivative of the log-pre-likelihood. On the interior $\theta > x_1$ we have $\ell_{x_1}(\theta) = -\ln\theta$, so

$$s_{x_1}(\theta) \;=\; \frac{\partial}{\partial\theta}\ell_{x_1}(\theta) \;=\; -\frac{1}{\theta}.$$

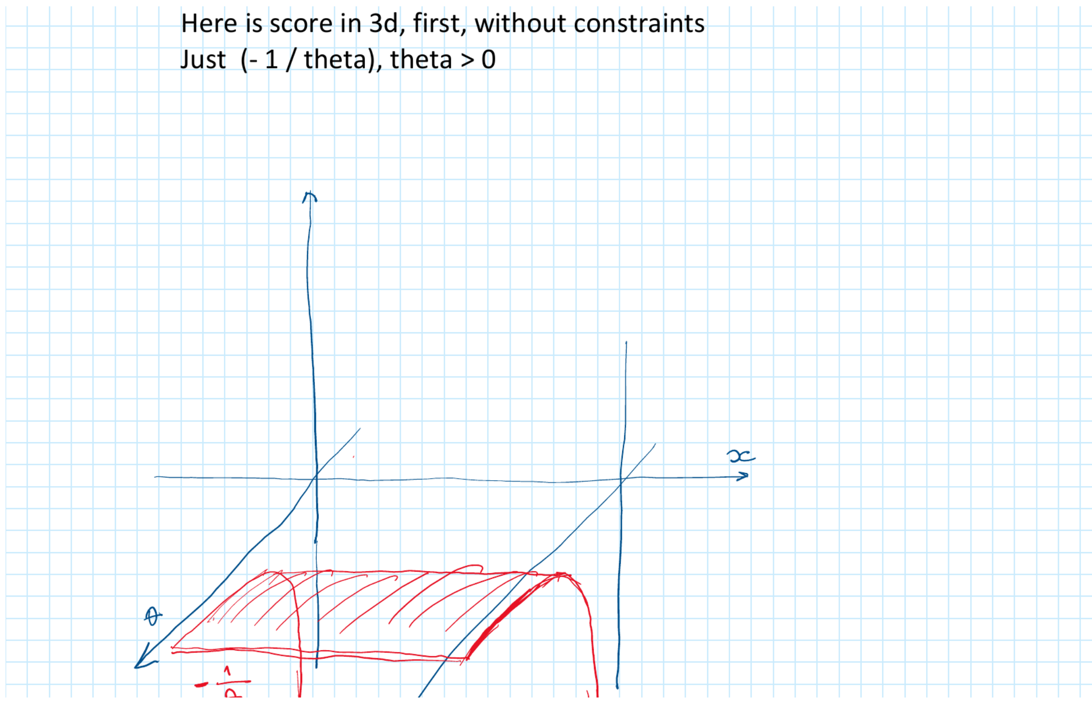
For $\theta < x_1$ the log-pre-likelihood is $-\infty$ (constant), so the derivative is zero by convention. At $\theta = x_1$ itself there is a jump discontinuity, so the derivative is undefined. Formally:

$$s_{x_1}(\theta) \;=\; \begin{cases} 0 & \text{by convention, } \theta < x_1, \\ \text{undefined} & \theta = x_1, \\ -1/\theta & \theta > x_1. \end{cases}$$



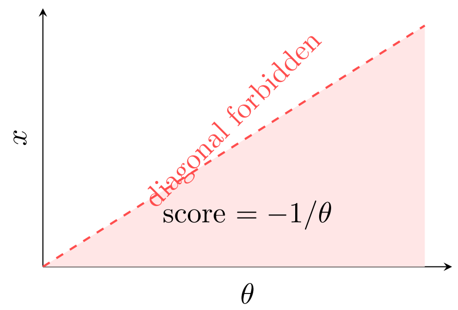
Key observation: at $\theta^* = x_1$ (the MLE), the pre-score is *not* zero — it jumps from 0 (left) to $-1/x_1$ (right). This is the hallmark of a non-regular family: the first-order condition $s = 0$ fails to locate the MLE.

Score in 3D. The bivariate pre-score $s(x, \theta) = -1/\theta$ (ignoring the constraint) is a surface that depends only on θ: flat in x, negative for all $\theta > 0$, increasing toward zero as $\theta \to \infty$. Without the constraint it is a flat red slab below the (x, θ)-plane:

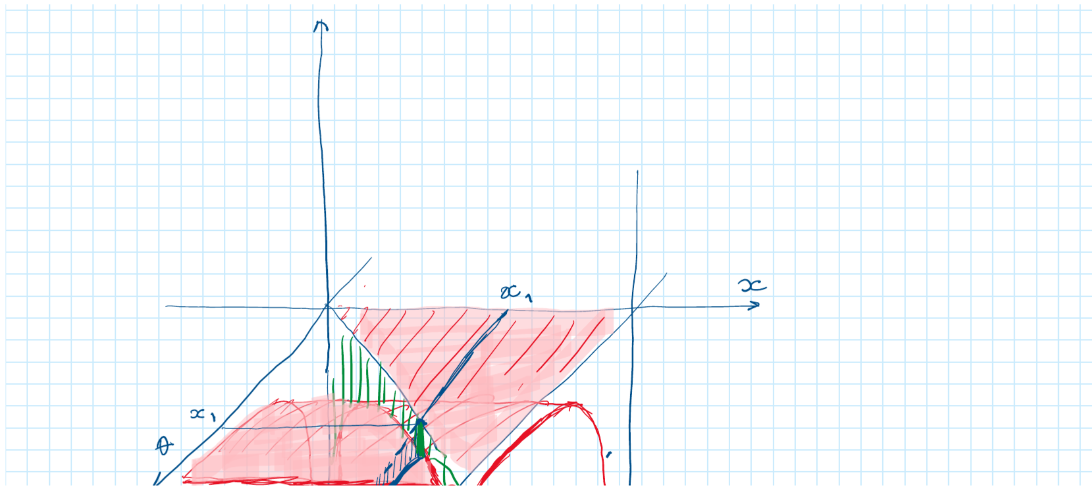


The constraint $0 < x < \theta$ is slightly more restrictive for the score than for the log-likelihood: the diagonal $x = \theta$ itself is *forbidden* (the density is zero there, so the score is undefined on the boundary), not merely a cliff edge. The domain $0 < x < \theta$ (strict inequalities) is the open triangular wedge:

Score domain: $0 < x < \theta$ **(diagonal** $x = \theta$ **forbidden)**



Adding the constraint (but before cutting with x_1) and then adding the x_1-section gives the full picture:

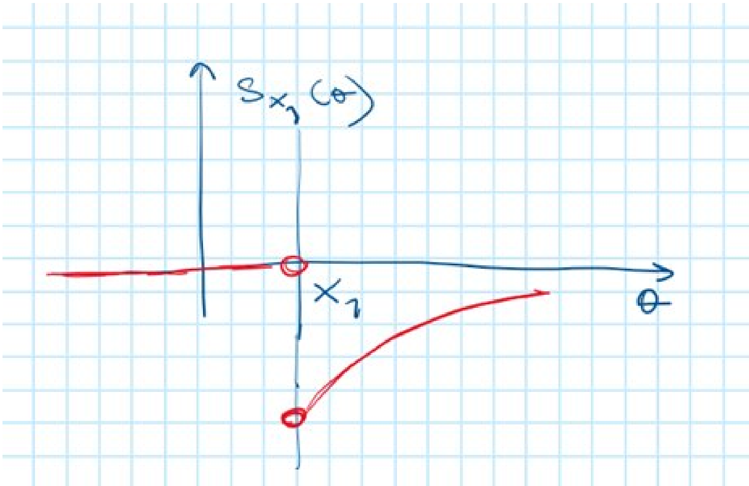


The x_1-section (blue curtain) shows the pre-score $s_{x_1}(\theta)$: zero for $\theta < x_1$, then jumping to $-1/x_1$ and decreasing as $-1/\theta$ for $\theta > x_1$.

1.11 Step 8. Score for n observations

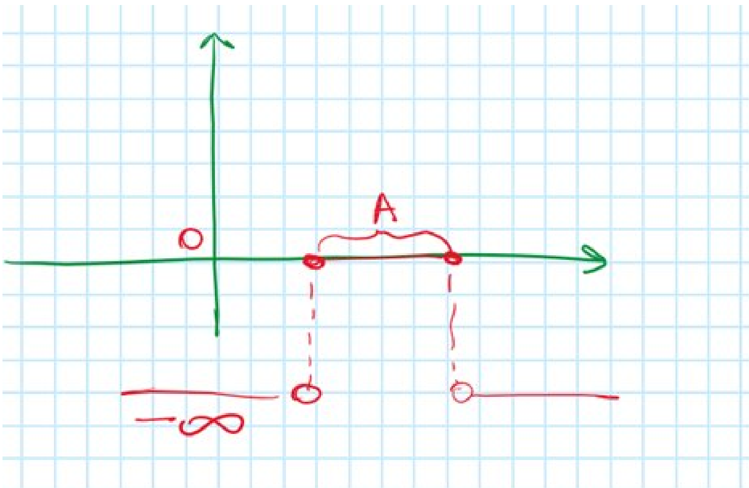
Recall: pre-score for $n = 1$. For a single observation the pre-score $s_{x_1}(\theta)$ is zero (by convention) left of x_1, undefined at x_1, and $-1/\theta$ to the right:

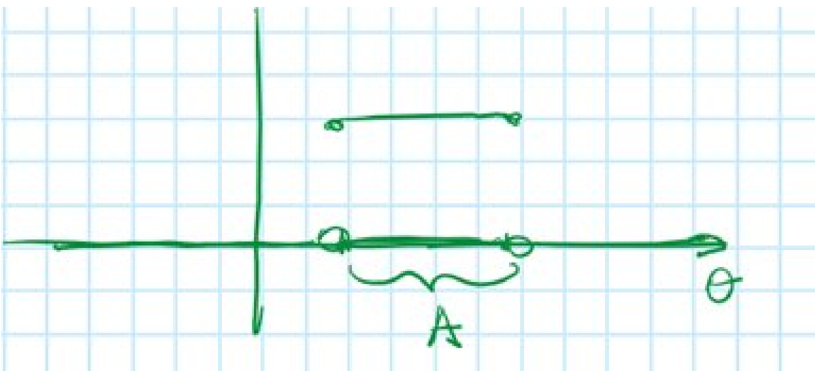
$$s_{x_1}(\theta) = \begin{cases} 0 & \theta < x_1, \\ \text{undefined} & \theta = x_1, \\ -1/\theta & \theta > x_1. \end{cases}$$



How to logarithm an indicator. A key step in passing from the likelihood to the log-likelihood for n observations is handling $\ln \mathbf{1}_A(\theta)$. The indicator $\mathbf{1}_A$ is 1 on A and 0 off A; its logarithm is 0 on A and $-\infty$ off A:

$$\ln \mathbf{1}_A(\theta) \;=\; \begin{cases} 0 & \theta \in A, \\ -\infty & \theta \notin A. \end{cases}$$





The product $\prod_i \frac{1}{\theta} \mathbf{1}_{x_i \le \theta}$ collapses under the log to

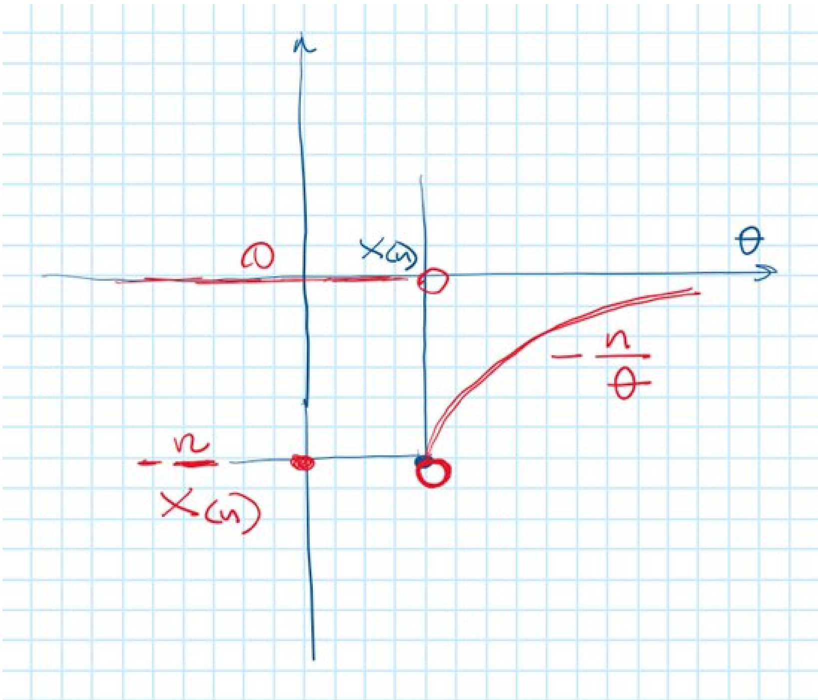
$$\ell_{x_1,\ldots,x_n}(\theta) \;=\; \ln \frac{1}{\theta^n} + \ln \mathbf{1}_{x_{(n)} \le \theta} \;=\; -n \ln \theta + 0 \;=\; -n \ln \theta, \qquad \theta \ge x_{(n)}.$$

Score for n observations. Differentiating gives:

$$s(\mathbf{x},\theta) \;=\; \begin{cases} -n/\theta & \theta > x_{(n)}, \\ \text{undefined} & \theta = x_{(n)}, \\ 0 \text{ (by convention)} & \theta < x_{(n)}. \end{cases}$$

Note that where the score is defined, it equals the sum of the individual pre-scores: $s_{x_1,\ldots,x_n}(\theta) = s_{x_1}(\theta) + \cdots + s_{x_n}(\theta)$, valid on their common domain $\theta > x_{(n)}$.

The 2D graph of s as a function of θ (analogous to the $n = 1$ picture above, but with $x_{(n)}$ as the cliff and $-n/\theta$ on the right):



Support, kernel, and where the cliff lives. Viewed as a function of θ alone, the score has two regions:

- **Support region** $\theta > x_{(n)}$: score $= -n/\theta$. It does not depend on $\mathbf{x}$ at all — only on θ.
- **Kernel region** $\theta < x_{(n)}$: score $= 0$ by convention. It also does not depend on the sample — it is identically zero regardless of which $\mathbf{x}$ was observed.

In both regions the score is constant as a function of $\mathbf{x}$. Yet the two regions are not the same: *the boundary between them — the cliff — depends on the sample through* $x_{(n)} = \max_i x_i$. The data determines where the jump is, not what the score value is on either side of it.

Score as a map: setting up the expectation. To compute $\mathbb{E}_\theta[s(\mathbf{X},\theta)]$ we have to be precise about what "expectation" means here. The score is a *map*:

$$s(\,\cdot\,,\theta)\colon \mathbf{x} \;\longmapsto\; s(\mathbf{x},\theta).$$

Taking its expectation means: generate the sample $\mathbf{X}$ from the same θ that appears in the score formula, then push the measure forward through this map. The map and the measure it acts on must be in agreement — we use one and the same θ for both.

Under $\mathrm{U}[0,\theta]$, each X_i lies in the open interval $(0,\theta)$ with probability one. Therefore $x_{(n)} < \theta$ almost surely, and the realised sample falls in the support region with probability one:

$$X_i \in (0,\theta) \;\text{ a.s.}$$

Not centered, and drifting farther from zero with n. In the support region the score equals $-n/\theta$ — constant in $\mathbf{x}$, and still less dependent on ω. The expectation is therefore

$$\mathbb{E}_\theta\big[s(\mathbf{X},\theta)\big] \;=\; \int_{[0,\theta]^n} \Big(-\frac{n}{\theta}\Big)\,\frac{1}{\theta^n}\,d\mathbf{x} \;=\; -\frac{n}{\theta}.$$

The score identity $\mathbb{E}[s] = 0$ fails completely, and the magnitude n/θ grows without bound as $n \to \infty$. Far from converging to zero, the expected score drifts farther from it with every additional observation.

The value 0 in the kernel region is a *conventional* assignment — the derivative of the constant $-\infty$ — not a value that the score actually takes when the sample is generated from θ. Zero is never the true computed score; it only appears as a bookkeeping convention at the boundary and beyond.

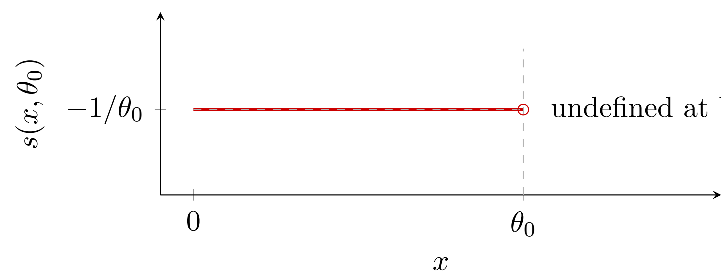
Finally, the score at the MLE is *undefined*: $\hat\theta = x_{(n)}$ is exactly the cliff point where the log-likelihood has a jump discontinuity and the derivative does not exist. The first-order condition $s = 0$, which locates the MLE in regular families, has no counterpart here.

1.12 Step 7. Score and the failure of regularity

On the interior of the support $x \in (0,\theta)$, where $h > 0$, the pre-score is

$$s(x,\theta) \;=\; \frac{\partial}{\partial\theta}\ln h(x,\theta) \;=\; \frac{\partial}{\partial\theta}\ln\frac{1}{\theta} \;=\; \frac{\partial}{\partial\theta}(-\ln\theta) \;=\; -\frac{1}{\theta}.$$

Pre-score $s(x,\theta_0) = -1/\theta_0$: constant in x on $(0,\theta_0)$



Three things go wrong simultaneously.

- **Score identity fails.** The score $s(x,\theta) = -1/\theta$ does not depend on x at all. Its expectation under $\mathrm{U}[0,\theta]$ is
$$\mathbb{E}_\theta[s(X,\theta)] \;=\; \int_0^\theta \Big(-\frac{1}{\theta}\Big)\,\frac{1}{\theta}\,dx \;=\; -\frac{1}{\theta} \;\neq\; 0.$$
The score identity $\mathbb{E}[s] = 0$ fails because the support boundary $x = \theta$ depends on θ; differentiating under the integral sign (Leibniz rule) is not valid here.

- **Score is degenerate.** Since $s(x,\theta) = -1/\theta$ is constant in x, it is not a random variable in any useful sense: it carries no information about the specific realisation. Its variance is zero, so the Fisher information $\mathcal{I}(\theta) = \mathrm{Var}_\theta[s] = 0$ by the usual formula. (The true information is finite but must be computed differently for non-regular families.)

- **MLE is not found by score $= 0$.** The log-likelihood $\ell(\theta) = -n\ln\theta$ has derivative $-n/\theta \neq 0$ everywhere on its support $\big[x_{(n)}, \infty\big)$. The maximum is at the *left boundary* $\hat\theta = x_{(n)}$, not at an interior critical point. Score-based first-order conditions are inapplicable.

CLT for score does not apply. In the normal example, the standardised score $S_n/\sqrt{n\mathcal{I}(\theta_0)} \xrightarrow{d} \mathcal{N}(0,1)$ because the score is centred ($\mathbb{E}[s]=0$) and has finite variance. For $\mathrm{U}[0,\theta]$ neither condition holds. The sample maximum $x_{(n)}$ converges to θ_0 at rate n (not $\sqrt{n}$, not $\log n$): one verifies directly that for $z \geq 0$

$$P\big(n(\theta_0 - X_{(n)}) > z\big) \;=\; \Big(1 - \tfrac{z}{n\theta_0}\Big)^n \;\longrightarrow\; e^{-z/\theta_0},$$

so $n(\theta_0 - X_{(n)}) \xrightarrow{d} \mathrm{Exp}(1/\theta_0)$ — an Exponential distribution (the reversed Weibull extreme-value distribution with shape parameter 1). Compare: for $X_i \sim \mathrm{Exp}(\lambda)$ the normalising sequence for the maximum is $\log n$, not n, because that distribution has no finite right endpoint. The score-based Rao (Lagrange multiplier) test has no counterpart in this family.

Summary.

	$\mathcal{N}(\mu,1)$	$\mathrm{U}[0,\theta]$
MLE	$\bar{x}$ (score $=0$)	$x_{(n)}$ (boundary)
Score s	$x-\mu$, random in x	$-1/\theta$, constant in x
$\mathbb{E}[s]$	0 (regular)	$-1/\theta \neq 0$ (non-regular)
$\mathrm{Var}[s]$	1 (Fisher info)	0 (degenerate)
Rate of $\hat{\theta}$	$1/\sqrt{n}$	$1/n$
CLT for score	Yes (Wald, LR, Rao)	No